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Simulation of Air Pollution in Almaty City under Adverse Weather Conditions

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Abstract

The problem of air pollution in Almaty city in stagnant weather conditions is considered. Such conditions are so frequent, because the city is located at the foothills of the mountain range of Ile Alatau. The WRF-Chem complex model is used for the analysis and allows a high-detail simulation both of the atmospheric circulation and the transport of various pollutants, taking into account their chemical transformations. The results of modeling meteorological processes are compared with observational data. The main attention is paid to the calculation of the sulfur dioxide plume emitted by the pipes of the coal-fired CHP-2. The simulations showed that regional latitudinal wind, whose direction regularly changes to the opposite one, dominates in the northern part of the city atmosphere. During such periods and under unstable atmosphere stratification conditions, the plume

dissipates and pollutants penetrate into the surface layer. In some areas, the concentration of SO_2 significantly exceeds sanitary standards.

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1. INTRODUCTION

Almaty city ($43^{\circ}15' \text{ N}$, $76^{\circ}54' \text{ E}$) is the largest settlement in the Republic of Kazakhstan (the total area is $> 680 \text{ km}^2$), the population is $> 1.9 \cdot 10^6$ people. Most of the city, where the main part of the population and pollution sources are concentrated, is located on a relatively flat surface with a slope of $1^{\circ} - 2^{\circ}$ in the northern direction. The mountain regions, which have recently been joined to the city and are intended for the construction of resorts, have specific microclimate, there is no problem of atmospheric pollution.

Adverse ecological conditions in the atmosphere over the residential part of Almaty are associated with its location in the wind shadow of the mountain range of Ile Alatau, where stagnant weather conditions are frequent. In these conditions, the only ventilating factor is local winds resulting from orographic and thermal surface inhomogeneities. The complex wind regime is formed when the mountain-valley circulation, the circulation around the urban heat island, slope winds, inversion layers, and other processes interact with each other. Unfortunately, local winds cannot intensively transport pollutants out of the city air basin due to low wind speeds ($1 - 2 \text{ m/s}$). Calm weather and strong inversion layers suppressing the vertical exchange are the main causes for the high level of pollution in the atmosphere over the city. Similar conditions are registered in many industrial cities of the Russian Federation (Krasnoyarsk, Chita, etc.) located in the hollows of highly rough terrain. The scenario-based analysis of atmospheric processes in the mentioned cities was successfully carried out using the model developed at the Institute of Computational Mathematics and Mathematical Geophysics (Siberian Branch, Russian Academy of Sciences) [8, 9]. Probably, the experience of this work will be useful for modeling real episodes of high levels of atmospheric pollution in these cities.

The present study considers the application of the WRF-Chem (Weather Research and Forecasting with the Chemistry module) model [13] to the analysis of atmospheric processes in the autumn-winter season, when ecologically unfavorable synoptic

conditions of the Siberian High and low-gradient high- and low-pressure fields are realized over Almaty [7]. The CHP-2 combined heat and power plant was chosen as a modeling object, that operates at full capacity in this season and emits the great amount of pollutants. All simulations were performed for sulfur dioxide (SO_2), which is the most typical waste gas for the coal-fired enterprises.

The WRF-Chem research project [21] (<https://ruc.noaa.gov/wrf/wrf-chem/>) unites a large group of scientists who developed a powerful numerical method and a package of methods for the parameterization of physical and chemical processes. The parameterization of physical processes (turbulence, radiation, surface layer, etc.) is basically related to solving meteorological problems [2]. For the problem of transport and chemical transformation of pollutants, there is an option of using different mechanisms of photochemical reactions.

The CHP-2 emissions have a significant but not key influence on the ecological state of the atmosphere over Almaty. There are more than 10 heat and power plants within the city. According to the latest data (<https://vlast.kz/gorod/41241-tec-2-perevedut-na-gaz-k-2024-godu-no-eto-ne-tocno.html>), their emissions make up to 34% of the total volume, the contribution of motor transport exhaust gases is 52% and that of the private-sector stove heating is 14%. In the future, it is planned to apply the WRF-Chem to the analysis of the impact of all sources on the atmosphere state in different parts of the city, including the southern one, where the mountain-valley circulation and slope winds dominate.

The formation of the package of parameterization methods is a difficult problem, especially for the atmosphere of the city located in the mountain region. The choice of the optimum parameterization of physical processes was based on the statistically minimum deviation of simulation results from observations at weather stations in Almaty and its outskirts.

It should be noted that the successful application of the WRF-Chem depends on the availability of reliable data of atmospheric monitoring and data on pollution sources. In view of this, the World Meteorological Organization (WMO) launched the Integrated Urban Services (IUS) project, that was joined by a number of cities [10].

The present study is the continuation of the earlier studies [4, 5], where the simpler models (ModAP, HYSPLIT) were applied, and the relevant task of mapping the risks of air pollution over Almaty was set. Unlike these models, the WRF–Chem takes into account the impact of both the underlying surface and photochemical reactions on meteorological processes, which is quite significant when analyzing the pollution spread in the stagnant atmosphere.

2. DATA AND METHODS

2.1. Input Data for the WRF–Chem Model

Three domains (D1, D2, D3) in the Lambert conic projection in the Cartesian coordinate system with the center at the point of 43.24° N, 76.92° E were used for the WRF–Chem simulations. The first domain D1 of the regional scale covers the territory including Lake Balkhash in the north and the Tian Shan in the south. The second nested domain D2 of meso–meteorological scale includes Issyk–Kul Lake and the Ile Alatau range. The third nested domain D3 of local scale focuses on Almaty and its outskirts. The D1, D2, and D3 grids have a linear horizontal resolution of 9, 3, and 1 km, respectively. Each of the three domains contains 100 grid points from west to east and 100 points from south to north. Along the vertical, 38 levels are used, including 14 levels in the lower 200–meter layer. The time step for the D1, D2, D3 domains was 18, 6, and 2 s, respectively. The larger time steps (45, 15, and 5 s) were reduced to the unstable run. The instability was manifested during the computation in the last model domain, where complex mountain terrain was specified in detail based on the Aster GDEM three–dimensional terrain model (the resolution is 30 m).

The layer taking into account the land use was formed using the cartographic base of the city (streets, buildings, parks, reservoirs, and enterprises). To the standard set USGS (United States Geological Survey) (24 categories) used in the WRF–Chem by default, three categories of urban building were added: 31, 32, and 33 [17], which allowed including the BEP (Building Effect Parameterization) multi–layer scheme of urban canopy.

Geographic statistical WRF data specified by default (https://www2.mmm.ucar.edu/wrf/users/download/get_sources_wps_geog.htm) were replaced by the above land–use and terrain layers using the utility convert geotiff and were converted to the geogrid format.

The coordinates of CHP-2 pipes together with the information about their height and sulfur dioxide emission rates were obtained from the inventory of stationary sources from the existing projects of maximum permissible emissions. Data on the total annual emission were converted to specific power (g/s) under assumption of a constantly and uniformly operating source.

The choice of the characteristic (in terms of atmospheric pollution) period for modeling is of great importance. In winter, calm and low-wind conditions in Almaty and its outskirts mainly occur under the following synoptic conditions: the southern and southwestern peripheries of the Siberian (Asian) High wedge and the low-gradient high- or low-pressure field (<https://legalacts.egov.kz/npa/view?id=2017914>). It is natural that the concentration of pollutants reaches extremely high values during these periods.

To choose typical meteorological conditions, the synoptic charts for 2015–2017 were analyzed; the period from November 27 to December 4, 2016, during which three adverse (in terms of atmospheric pollution) synoptic regimes were registered, was taken for simulations. During the first five days (November 27 to December 1), the “Siberian High wedge” synoptic conditions were kept, which were replaced by the low-gradient high- (December 2) and low-pressure (December 3) field with light rains at the end of the period (December 3–4) as a result of the warm air invasion. It was taken into account that it was possible to test the WRF-Chem model under different weather conditions during this period. It should be noted that the first day of simulations (November 27) was removed from consideration, as it was assumed that the model “spin-up” and the formation of pollutant concentration fields take place during the first day.

Initial and boundary conditions were formed of ERA5 reanalysis data with a spatial resolution of 0.25° provided by the ECMWF. The variant with the real initial data for the outer domain was used, and, hence, the prescribed lateral boundary conditions were formed by the algorithm described in [21]. The boundary conditions for the nested domains were interpolated from the outer domain at each time step; the feedback was also realized (the parameter feedback = 1).

The comparative analysis of simulation results with ground-based observational data was performed using the information from the database of the University of Wyoming (<http://weather.uwyo.edu/upperair/sounding.html>), that contains sounding data, and the

information from the “World Weather” database (<http://rp5.kz/docs/about/ru>), that presents observational data from weather stations in Almaty and its outskirts.

2.2. Selecting Parameterization Methods in WRF–Chem Model

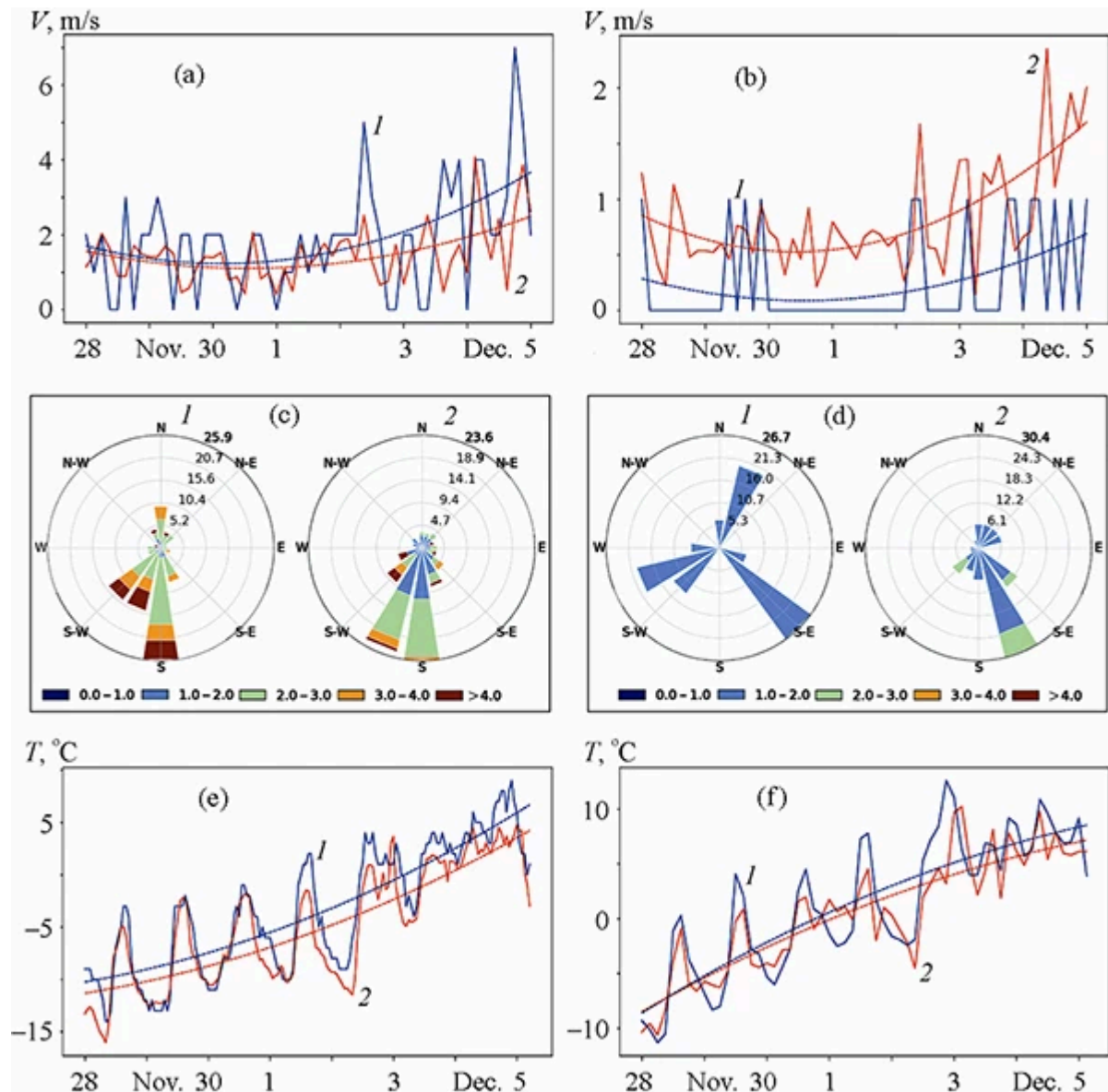
The problem of selecting the WRF parameterization for the adequate description of meteorological processes in the air basin over Almaty was solved using the studies [[6](#), [11](#), [19](#)] implemented for the similar complex territories. To choose the optimum configuration, more than 80 test simulations were used, the mean and absolute root-mean-square errors of temperature, wind speed, and wind direction simulation were calculated, and the test simulation with the minimum total error was chosen. Observational data from eight weather stations in Almaty and its outskirts were used to calculate the errors.

Thus, the variant with the following basic parameterization methods was chosen: the modified Mlawer’s parameterization for short- and long-wave radiation [[14](#)], the Bougeault–Lacarrere scheme for the boundary layer [[12](#)], the Kain–Fritsch parameterization for convection [[15](#)] for the domains D1 and D2, the multi-layer model of urban canopy for the urban territory [[16](#)], the Smagorinskii scheme for turbulence (horizontal diffusion) [[21](#)].

When setting up the chemical block, several variants of gas-phase chemical mechanisms were tested, and the RADM2 (Regional Acid Deposition Model, 2nd generation) variant, that is the most cost-effective in terms of computer time, was chosen [[22](#)].

3. ANALYSIS OF RESULTS

The comparison of simulation results with the observations of wind speed and temperature was carried out for Aeroport and Almaty weather stations (Fig. [1](#)). The regression curves constructed by the least-squares method are also drawn on the graphs of temporal variations in wind speed and temperature (Figs. [1a](#), [1b](#), [1e](#), and [1f](#)). The comparison shows that at Aeroport station (Figs. [1a](#) and [1c](#)), both simulation and observation data characterize wind as light, variable, and occasionally gusty. The wind speed variations are registered during the whole simulation period, but measurement data have a greater amplitude (0 to 3 m/s) than model data (0 to 2 m/s). The regression curves also indicate this fact, the curve for measurement data is situated over the simulated curve.

Fig. 1

The comparison of simulation results with observations at (a, c, e) Aeroport and (b, d, f) Almaty weather stations: (a, b) 10-m wind speed, (c, d) wind rose, and (e, f) 2-m air temperature for the period from 00:00 on November 28 to 05:00 on December 5, 2016. (1) Observations; (2) WRF simulations. The dotted line is the second-order regression curves constructed by the least-squares method.

The comparison of observations at Almaty station and simulation results is presented in Fig. 1b. Unfortunately, the procedure for rounding measurement data to the integer values reduced to 0 and 1 m/s the entire spectrum of wind speed variations in the range from 0 to 1.5 m/s. In view of this, the main conclusions were based on the regression analysis data. It is clear that according to the observations and simulations, the anticyclonic period is

characterized by low wind speed, but the model overestimates it on average by 0.5 m/s. In case of warm air invasion started on December 2, wind strengthens on average, based both on observations and simulations but the difference increases to 1 m/s. Thus, model wind speeds are on average higher than the observed ones by 0.5–1.0 m/s. Probably, the reason is that light local winds are decelerated by the buildings located near Almaty weather station. The modeling of such urban effects requires calculations on a denser grid or more accurate consideration of urban building effects [18].

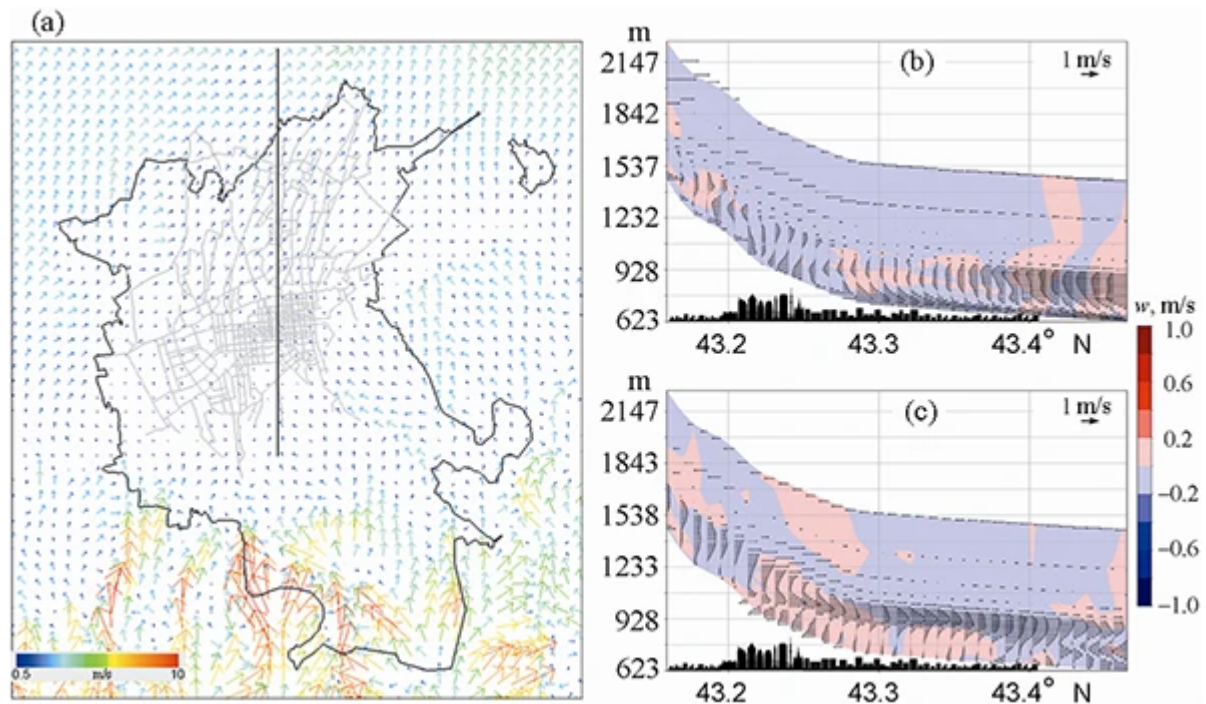
For the comparative analysis of wind direction, wind roses were constructed (Figs. 1c, 1d). The substantial difference between the diagrams based on observational data from Almaty and Aeroport stations is associated with the effect of urban building. At Almaty station, calm conditions were registered in 75% of cases. The other 25% are not sufficient to determine a prevalent wind direction. According to model simulations, the wind from mountains has the maximum frequency (Fig. 1d).

The observations at Aeroport weather station were more informative for constructing the wind rose: the measurements were more frequent (every half an hour), and the number of cases with absolutely calm conditions was smaller, since the airflow was not decelerated by surrounding buildings. The data of observations and simulation in the area of this weather station (Fig. 1c) coincide well in the prevalent wind direction: the wind from mountains (katabatic wind) dominates in both diagrams, which is natural for the autumn–winter season. In terms of wind power, the model wind is weaker than the observed one, probably caused by the fact that the model grid cell (1×1 km) covers the airport buildings and a part of residential buildings.

The model results clearly demonstrate the wind distribution in horizontal (surface) and vertical planes (Fig. 2). The surface wind field at 00:00 (Fig. 2a) and vertical sections at 00:00 and 12:00 (Figs. 2b and 2c) local time under anticyclonic conditions (November 29, 2016) are given. As expected, in the mountains, the complex pattern of wind speed with its significant drops is observed, and the residential urban territory (it was represented by the street–road network) is characterized by low wind speeds and local calm conditions (Fig. 2a). The low–wind surface field is also observed in the daytime, but with a considerable difference in the structure of flows, which is clearly manifested at the vertical sections of the wind field (Figs. 2b and 2c). In these figures, the projection of the wind speed vector to the south–north axis passing through the location of Almaty weather station was drawn to distinguish mountain–valley winds. As clear from the figures, the

airflow structure clearly indicates different phases of the mountain–valley circulation: the nighttime phase, when cold downslope winds are registered (Fig. 2b), and the daytime movement of warm air up the slope (Fig. 2c). The counterflows are formed in the upper layers of the urban atmosphere. As expected, the anabatic wind has a smaller speed than the katabatic one.

Fig. 2



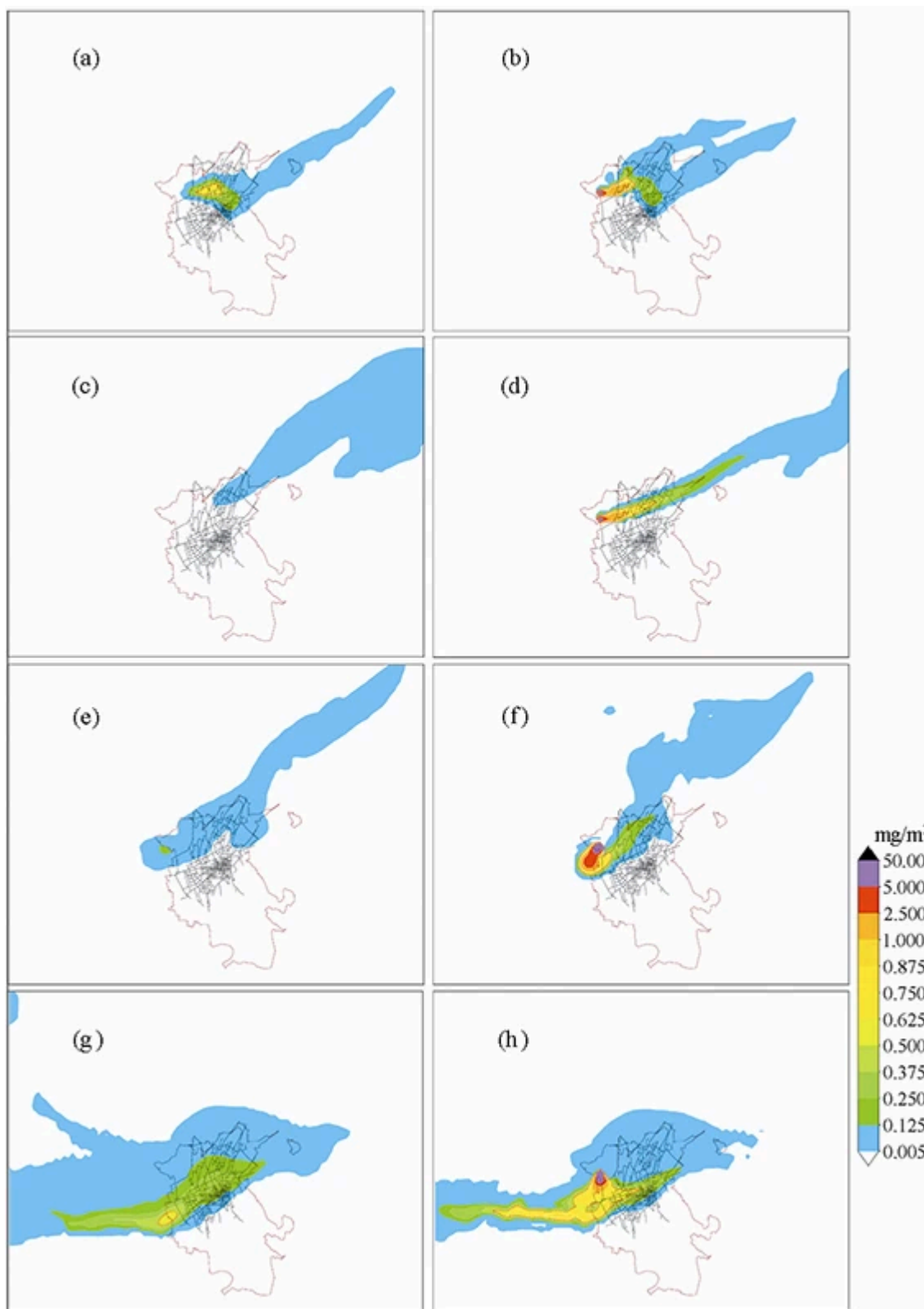
The wind field over Almaty on November 29, 2016: (a) the surface 10-m wind field at 00:00; vertical sections of the wind field at (b) the nighttime (00:00) and (c) daytime (12:00) moments.

The results confirm the forecasts of scientists [1] that it is promising to use the numerical analysis of mesoscale and local processes in the areas where the impact of large-scale flows is minimal as compared to the processes generated by surface inhomogeneities. The consistency of model simulations with measurement data is registered in the piedmont regions, and the approximation errors significantly increase in the mountains, where intermittent, sometimes strong wind is observed. A good predictability of wind and temperature regimes in the cross-country areas was noted in [11, 20, 23].

The complex pattern of wind in the air basin over Almaty under stagnant conditions is more strongly manifested when analyzing the transport of pollutants. Figure 3 presents

the results of simulation of the SO₂ plume from the CHP-2 source located in the northern part of Almaty for different time moments at the first (0 to 14 m) and at the ninth (119–132 m) simulation levels, that correspond to the height of the CHP-2 pipe (129 m). Unfortunately, data needed to calculate the effective height of the pipe (temperature and rate of released gases, the pipe diameter) are not available to the public. The presented fields of SO₂ concentration indicate the most typical situations. The analysis of pollution dynamics demonstrated that airflows in the northern part of the city form two regimes of the spread of plumes of the released substances: from west to east and from east to west. These sublatitudinal flows are the display of regional processes close to the mountain ranges. Western wind prevailed in the analyzed period under anticyclonic conditions (November 28–30); the daytime pollutant plume was unstable and reached the surface layer (Figs. [3a](#) and [3b](#)). However, during most of the day (in the evening, at night, and early in the morning), the plume had a form of a narrow jet and spread for large distances with the minimum contamination of the surface layer (Figs. [3c](#) and [3d](#)). This probably depends on the atmospheric stratification: in the evening, the katabatic wind cools down the surface layer and creates conditions for the stable atmosphere; in the daytime, the ground is heated, the inversion layer ascends above the pipe, and the plume enters the unstable atmosphere. Strong pollution of the surface layer was also registered during the change in the direction of the latitudinal transport. Figures 3e and 3f present an example of the change of western wind to eastern. This regime usually takes place in the morning after the sunrise, and citizens observe a dirty air layer in the north at that time. In the daytime, due to the anabatic lifting, the polluted air moves towards the mountains. Over the entire simulation period, such regime was most strongly pronounced on December 2 (Figs. [3j](#) and [3h](#)), when synoptic conditions of the low-gradient high-pressure field were observed. In the morning on that day, two processes matched: the transformation of the latitudinal transport from the eastern to western directions and the wind strengthening toward the mountains, that was caused by the surface heating in the piedmont regions as a result of warm air arrival. The simulation results showed the highest level of pollution of the atmospheric surface layer over the city on that day, locally above the maximum single permissible concentration ($MPC_{m.s}$).

Fig. 3



The fields of atmospheric pollution with sulfur dioxide SO_2 from the CHP-2 source in Almaty at the levels of (a, c, e, g) 0–14 m and (b, d, f, h) 119–132 m in the autumn–winter season of 2016: at (a, b) 16:00 and (c, d) 22:00 on November 29, (e, f) at 10:30 on November 30, and (g, h) at 10:00 on December 2, 2016.

Unfortunately, historical data from air pollution observation stations in Almaty are not available to the public. The monthly information bulletins on environmental conditions

are published (<https://www.kazhydromet.kz/ecology/informacionnye-byulleteni-o-sostoyanii-okruzhayuschey-sredy-respubliki-kazahstan>). According to this information, the maximum concentration of SO₂ in November 2016 was equal to 0.568 mg/m³ (1.135 MPC_{m.s}), which corresponds (within the margin of error) to the result of model simulations (0.387 mg/m³), but average daily concentrations of SO₂ in some areas in the northern part of the city, unlike the bulletin data, are much higher than the average daily MPC (this normal is equal to 0.05 mg/m³).

It is noteworthy that a different wind regime is realized in the southern part of the city during the atmospheric stagnation. In that part, the effect of latitudinal transport on the pollution of urban environment is minimal, and mountain-valley and slope flows prevail [3]. Their impact on the atmospheric pollution requires additional research.

4. CONCLUSIONS

It is shown that the WRF-Chem model adequately simulates the main processes in the atmosphere over Almaty under calm conditions if the relevant methods for the parameterizations of physical and chemical processes are chosen. The simulation results were compared with observations at two weather stations during the period from November 28 to December 4, 2016, that covers three ecologically unfavorable synoptic conditions: the Siberian High wedge, the low-gradient high- and low-pressure fields. The modeling revealed a complex, dynamic pattern of local winds with low speeds and different spatial distributions. The pattern of pollutant transport in these complex meteorological conditions was studied by the example of modeling the plume of SO₂ released from the pipes of CHP-2, that is the most intensive source of pollutants among all heat and power plants in Almaty. The simulations demonstrated that the dominating process in the northern part of the city is the regional sublatitudinal transport. The maximum level of pollution arises in case of the turbulent dissipation of the plume and of the regular transformation of the direction of latitudinal transport to the opposite one. It should be noted that the WRF-Chem model is applicable to the regions where large-scale flows exert smaller influence than local atmospheric processes (the mountain-valley circulation, flows around the urban heat island, etc.).

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Author information

Authors and Affiliations

EcoRisk, 050026, Almaty, Kazakhstan

E. A. Zakarin, L. A. Balakay, T. V. Dedova & K. A. Bostanbekov

Danish Meteorological Institute, 2100, Copenhagen, Denmark

A. A. Baklanov

Corresponding author

Correspondence to [T. V. Dedova](#).

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